

*Navier-Stokes equations in gas dynamics:
Green's function, singularity, and well-posedness*

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- Well-posedness of weak solution is a subtle subject.
- Well-Posedness may not valid for weak solution.

Non-uniqueness results for weak solutions of incompressible Euler equations, using the technique of convex integration:

Scheffer, V. An inviscid flow with compact support in space-time. J. Geom. Anal. 3(4), 343-401 (1993).

De Lellis C, Szekelyhidi L. On admissibility criteria for weak solutions of the Euler equations. Arch. Ration. Mech. Anal. 195 (2010), no. 1, 225-260.

- There well-posedness theories for system of hyperbolic conservation laws.

The existence theory is done by Glimm scheme:

Glimm, James. Solutions in the large for nonlinear hyperbolic systems of equations. Comm. Pure Appl. Math. 18 (1965), 697-715.

The continuous dependence on the initial data is done by the functional approach:

Liu, T.-P. The deterministic version of the Glimm scheme. Comm. Math. Phys. 57 (1977), no. 2, 135-148.

Liu, T.-P.; Yang, T. A new entropy functional for a scalar conservation law. Comm. Pure Appl. Math. 52 (1999), no. 11, 1427-1442.

By the homotopy approach: Bressan, A. Contractive metrics for nonlinear hyperbolic systems. Indiana Univ. Math. J. 37 (1988), no. 2, 409-421.

On the other hand, there is no proof of well-posedness theory for solutions obtained by the technique of compensated compactness. **The well-posedness theory is done within the Glimm solution algorithm.**

- Weak solution for compressible Navier-Stokes equation:

(No proof of well-posedness)

Hoff, D., Global solutions of the Navier-Stokes equations for multidimensional compressible flow with discontinuous initial data. JDE, 120 (1995) 215-254.

Lions, P.L., Mathematical Topics in Fluid Mechanics: Volume 2: Compressible Models. Oxford Lecture Series in Mathematics and its Applications, 10. (1996).

Feireisl, E. Dynamics of Viscous Compressible Fluids. Oxford Lecture Series in Mathematics and its Applications, 26. (2004).

- The purpose of this talk is to present another example of well-posedness theory for the weak solutions, besides that for system of hyperbolic conservation laws. We consider the simplest system of viscous conservation laws in physics, the Navier-Stokes equations in gas dynamics.

- Compressible NS equation in the Lagrangian coordinate

$$\begin{cases} v_t - u_x = 0, \\ u_t + p(v)_x = \kappa(u_x/v)_x, \\ p(v) = v^{-\gamma}, \gamma \in (1, 5/3), \kappa = 1, \end{cases} \quad \begin{cases} v : \text{volume per unit mass,} \\ u : \text{particle velocity,} \\ p(v) : \text{pressure with } \gamma \text{ law,} \\ \kappa : \text{Dissipation constant.} \end{cases}$$

- Nash, J., *Le probleme de Cauchy pour les equations differentielles d'un fluide general*, Bull. Soc. Math. France, 90 (1962) 487-497.
- Kanel, Ya . I., *On a model system of equations for one-dimensional gas motion*, Diff. Eq. (in Russian) 4 (1968) 721-734.
- Kazhikhov, A. V. and Shelukhin, V . V ., *Unique global solution in time of initial boundary value problems for one-dimensional equations of a viscous gas*, Prikl. Mat. Mech., 41 (1977) 282-291.
- Itaya , N., *On the Cauchy problem for th e system of fundamental equations describing the movement of compressible viscous fluid*, Kodai M ath. Sem . Rep., 23 (1971) 60-120.
- Matsumura, A., and Nishida, T., *The initial value problem for the equations of motion of viscous and heat conductive gases*. J. Math. Kyoto Univ. 20-1 (1980) 67-104.
- Hoff, D., *Global well-posedness of the Cauchy problem for the Navier-Stokes equations of nonisentropic flow with discontinuous initial data*, Journal of differential equations, JDE, 95 (1992) 33-74.
- Lions, P.L., *Mathematical Topics in Fluid Mechanics: Volume 2: Compressible Models*. Oxford Lecture Series in Mathematics and its Applications, 10. (1996).
- Feireisl, E. *Dynamics of Viscous Compressible Fluids*. Oxford Lecture Series in Mathematics and its Applications, 26. (2004).

Some remarks on weak solutions.

- NS.

$$\begin{cases} v_t - u_x = 0, \\ u_t + p(v)_x = \kappa(u_x/v)_x. \end{cases}$$

- The weak solution:

$$\begin{aligned} \int_0^\infty \int_{\mathbb{R}} -(\phi_t v - \phi_x u) dx dt &= 0, \\ \int_0^\infty \int_{\mathbb{R}} -(\psi_t u + \psi_x (p(v) - \kappa u_x/v)) dx dt &= 0 \text{ for all test functions } \phi, \psi. \end{aligned}$$

Remark: This formulation allows $v(x, t)$ and $u_x(x, t)$ to contain discontinuities; and the discontinuities can survive for $t > 0$ due to the hyperbolic-parabolic structure.

- Artificial viscosity coefficient problem:

$$\begin{cases} v_t - u_x = \kappa v_{xx}, \\ u_t + p(v)_x = \kappa u_{xx}. \end{cases}$$

- The weak solution:

$$\begin{aligned} \int_0^\infty \int_{\mathbb{R}} -(\phi_t v - \phi_x u + \kappa \phi_{xx} v) dx dt &= 0, \\ \int_0^\infty \int_{\mathbb{R}} \left(-(\psi_t u + \psi_x p(v)) - \kappa \psi_{xx} u \right) dx dt &= 0 \text{ for all test functions } \phi, \psi. \end{aligned}$$

Remark: $u(x, t)$ and $v(x, t)$ become continuous for $t > 0$.

Rankine-Hugoniot condition for discontinuities in NS

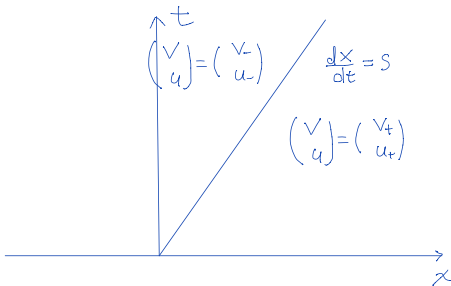
- NS in the conservative form.

$$\begin{pmatrix} v \\ u \end{pmatrix}_t + \begin{pmatrix} -u \\ p(v) - \kappa u_x / v \end{pmatrix}_x = \vec{0}.$$

- Rankine-Hugoniot Condition.

$$\begin{cases} s[v] = -[u], \\ s[u] = [p(v) - \kappa u_x], \\ \text{Continuity in } u \implies [u] = 0 \implies s = 0. \end{cases}$$

- All positions of discontinuities are stationary.



Itay's approach (Picard's iteration)

$$\begin{cases} \partial_t v^n - \partial_x u^n = 0, \\ \partial_t u^n - \partial_x \frac{\kappa}{v^{n-1}} \partial_x u^n = -\partial_x p(v^{n-1}), \text{ (linear heat equation in } u^n) \\ u^n(x, 0) = u_0(x), \quad v^n(x, 0) = v_0(x), \\ v_0 \in C^\sigma(\mathbb{R}), \quad u_0 \in C^{1+\sigma}(\mathbb{R}), \quad \sigma \in (0, 1). \end{cases}$$

Levi's parametrix method (regularity condition required on $v^{n-1}(x, t)$. This is ensured by the initial condition.)

$\Rightarrow \exists H(x, t; y, \tau; v^{n-1})$: The Fundamental solution of

$$\begin{cases} \left(\partial_t - \partial_x \frac{\kappa}{v^{n-1}(x, t)} \partial_x \right) H(x, t; y, \tau; v^{n-1}) = 0, \quad t \geq \tau, \\ H(x, \tau; y, \tau; v^{n-1}) = \delta(x - y). \end{cases}$$

$\Rightarrow$ Convergence of the iterations:

$$\begin{cases} u^n(x, t) = \int_{\mathbb{R}} H(x, t; y, 0; v^{n-1}) u_0(y) dy + \int_0^t \int_{\mathbb{R}} \partial_y H(x, t; y, \tau; v^{n-1}) p(v^{n-1}) dy d\tau, \\ v^n(x, t) = v_0(x) - \int_0^t \partial_x u^n(x, \tau) d\tau. \end{cases}$$

- Remark: This becomes a standard procedure to obtain the local existence theory in the L^2 energy estimates for to yield the **global stability** for solutions of the compressible Navier-Stokes developed by Matsumura-Nishida.

Functional space for well-posedness

Similar to Glimm's works on the hyperbolic conservation laws, one may consider the space of the **bounded variation (BV)** functions as a good candidate for the proper space to accommodate the weak solutions.

- Assumption. The initial data in the $BV \cap L^1$ class.
- **Missing component in the Picard's iteration: Precise pointwise and regularity structure of a fundamental solution $H(x, t; y, \tau; v^{n-1})$ when v^{n-1} is a BV function.**

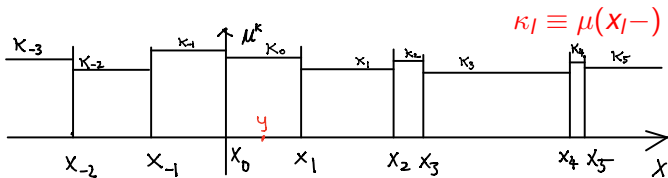
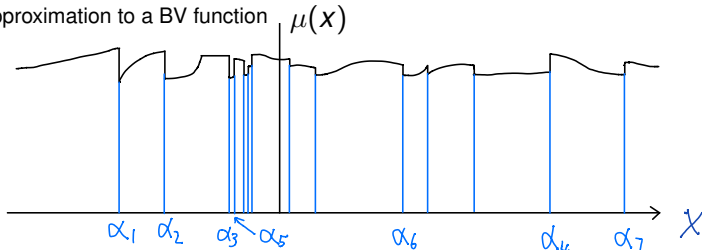
To construct the fundamental solution $H(x, t; y, \tau; \mu)$ of $\partial_t u - \partial_x \mu(x, t) \partial_x u = 0$ with precise pointwise classical description up to **2nd derivatives** with $\mu(x, t)$ in the **BV class**.

- The Levi's parametrix method can not applied due to lacking regularity property in $\mu(x, t)$.
- De Giorgi, Nash, Moser, etc. had the fundamental solution obtained by **Riesz' representation** and estimated in the Hölder continuities. It is **NOT SUFFICIENT** for the nonlinear problem.

A functional analysis for the heat equation

$$\partial_t u - \partial_x \mu \partial_x u = 0$$

- Approximation to a BV function $\mu(x)$



$$\inf_{j \in \mathbb{Z}} |x_j - x_{j-1}| > 0.$$

$$\lim_{k \rightarrow \infty} \|\mu - \mu^k\|_\infty = 0.$$

Combinatoric solutions

A procedure to obtain combinatoric the fundamental solution $H(x, t; y, \mu^k)$ in the weak sense:

$$\begin{cases} H(x, t; y; \mu^k) \equiv u(x, t), \\ \partial_t u - \partial_x(\mu^k(x) \partial_x u) = 0, \text{ (weak sense),} \\ u(x, 0) = \delta(x - y), \\ \mu^k(x): \text{ A step functions independent of } t. \end{cases}$$

• weak sense $\implies$

$$\begin{cases} u(x, t) : \text{ Continuous,} \\ \mu^k(x) u_x(x, t) : \text{ Continuous.} \end{cases}$$

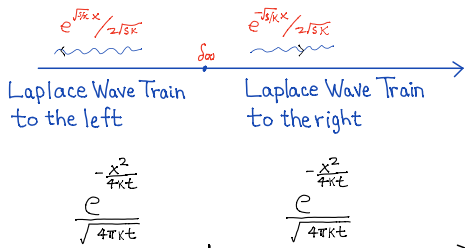
Laplace wave train

$$\begin{cases} \partial_t u = \kappa \partial_x^2 u, \\ u(x, 0) = \delta(x). \end{cases} \iff u(x, t) = \frac{e^{-\frac{x^2}{4\kappa t}}}{\sqrt{4\kappa\pi t}}$$

$$\mathbb{L}[u](x, s) \equiv \int_0^\infty e^{-st} u(x, t) dt. \text{ (Laplace Transform in } t)$$

$$s\mathbb{L}[u](x, s) - \kappa \partial_x^2 \mathbb{L}[u](x, s) = \delta(x) \iff$$

$$\mathbb{L}[u](x, s) = \begin{cases} \frac{e^{-\sqrt{\frac{s}{\kappa}}x}}{2\sqrt{s\kappa}}, & x > 0, \\ \frac{e^{\sqrt{\frac{s}{\kappa}}x}}{2\sqrt{s\kappa}}, & x < 0. \end{cases} \quad \text{Laplace wave train.}$$



$$\mathbb{L}^{-1}\left[\frac{e^{-\sqrt{\frac{s}{\kappa}}x}}{2\sqrt{s\kappa}}\right] = \frac{e^{-\frac{x^2}{4\kappa t}}}{\sqrt{4\kappa\pi t}}$$

Composite wave train: I

$$\begin{cases} \partial_t u = \partial_x \mu(x) \partial_x u, \\ u(x, 0) = \delta(x - y), \end{cases} \quad \mu(x) = \begin{cases} \kappa_-, x < 0, \\ \kappa_+, x > 0, \end{cases}$$

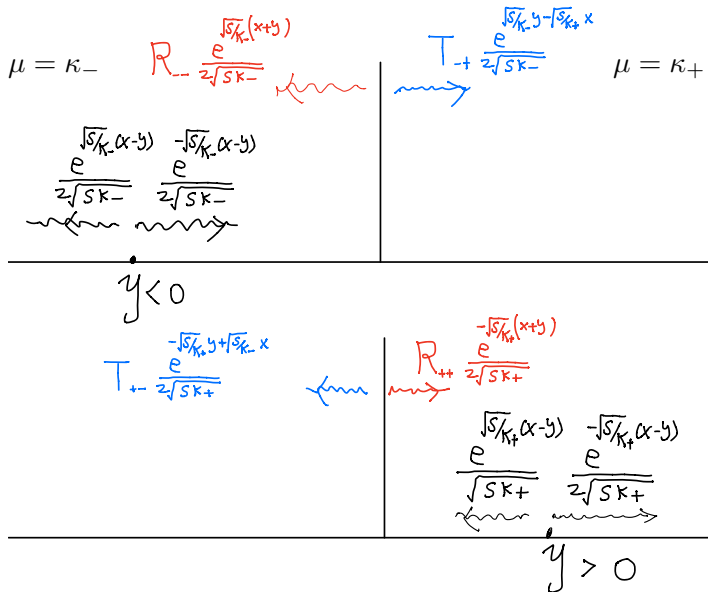
$$\mathbb{L}[u](x, s) = \begin{cases} \frac{e^{-\sqrt{s/\kappa_-}|x-y|}}{2\sqrt{s\kappa_-}} + R_{--} \frac{e^{\sqrt{\frac{s}{\kappa_-}}(x+y)}}{2\sqrt{s\kappa_-}}, & x, y < 0 \\ T_{-+} \frac{e^{\sqrt{s/\kappa_-}y - \sqrt{s/\kappa_+}x}}{2\sqrt{s\kappa_-}}, & x > 0, y < 0, \\ \frac{e^{-\sqrt{s/\kappa_+}|x-y|}}{2\sqrt{s\kappa_+}} + R_{++} \frac{e^{\sqrt{s/\kappa_+}(x+y)}}{2\sqrt{s\kappa_+}}, & x, y > 0 \\ T_{+-} \frac{e^{-\sqrt{s/\kappa_+}y + \sqrt{s/\kappa_-}x}}{2\sqrt{s\kappa_+}}, & x < 0, y > 0. \end{cases}$$

Weak solution $\implies \mathbb{L}[u](x, s)$ and $\mu(x)\partial_x \mathbb{L}[u](x, s)$: Continuous functions.

$$\begin{cases} T_{-+} = 1 + \frac{\sqrt{\kappa_+} - \sqrt{\kappa_-}}{\sqrt{\kappa_+} + \sqrt{\kappa_-}}, \\ R_{--} = \frac{\sqrt{\kappa_+} - \sqrt{\kappa_-}}{\sqrt{\kappa_+} + \sqrt{\kappa_-}}, \end{cases} \quad \begin{cases} T_{+-} = 1 - \frac{\sqrt{\kappa_+} - \sqrt{\kappa_-}}{\sqrt{\kappa_+} + \sqrt{\kappa_-}}, \\ R_{++} = -\frac{\sqrt{\kappa_+} - \sqrt{\kappa_-}}{\sqrt{\kappa_+} + \sqrt{\kappa_-}}. \end{cases}$$

$$T_{\pm\mp} = 1 + O(1)(\kappa_- - \kappa_+), \quad R_{\pm\pm} = O(1)(\kappa_- - \kappa_+) \text{ when } |\kappa_- - \kappa_+| \ll 1.$$

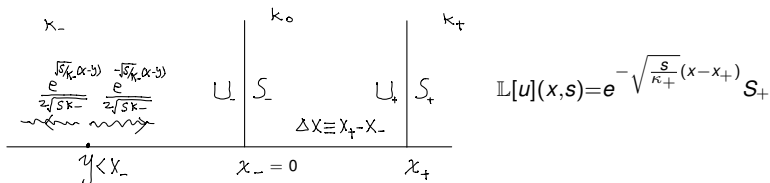
Diagram for composite wave train I.



Composite wave train II

$$\begin{cases} \partial_t u = \partial_x \mu(x) \partial_x u, \\ u(x, 0) = \delta(x - y), \end{cases} \quad \mu(x) = \begin{cases} \kappa_-, & x < x_-, \\ \kappa_0, & x \in (x_-, x_+), \\ \kappa_+, & x > x_+, \end{cases}$$

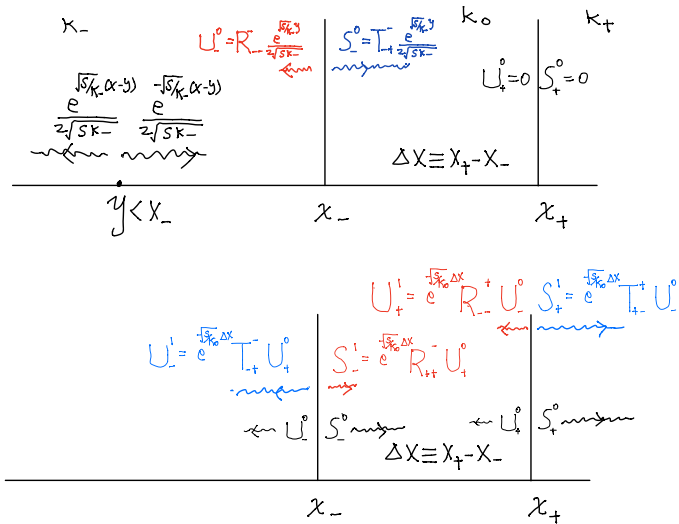
- Two diagrams for $y < x_- = 0$: I. (Resolvent Picture)



$$\begin{cases} U_- = R_{--}^- \frac{e^{\sqrt{s/\kappa_-}y}}{2\sqrt{s\kappa_-}} + T_{+-}^- e^{-\sqrt{s/\kappa_0}\Delta x} U_+, \\ S_- = T_{-+}^- \frac{e^{\sqrt{s/\kappa_-}y}}{2\sqrt{s\kappa_-}} + R_{++}^- e^{-\sqrt{s/\kappa_0}\Delta x} U_+, \end{cases} \quad \begin{cases} U_+ = R_{--}^+ e^{-\sqrt{s/\kappa_0}\Delta x} S_-, \\ S_+ = T_{-+}^+ e^{-\sqrt{s/\kappa_0}\Delta x} S_-. \end{cases}$$

$$\begin{pmatrix} U_- \\ S_- \\ U_+ \\ S_+ \end{pmatrix} = \begin{pmatrix} 1 & 0 & -T_{+-}^- e^{-\sqrt{s/\kappa_0}\Delta x} & 0 \\ 0 & 1 & -R_{++}^- e^{-\sqrt{s/\kappa_0}\Delta x} & 0 \\ 0 & -R_{--}^+ e^{-\sqrt{s/\kappa_0}\Delta x} & 1 & 0 \\ 0 & -T_{-+}^+ e^{-\sqrt{s/\kappa_0}\Delta x} & 0 & 1 \end{pmatrix}^{-1} \begin{pmatrix} R_{--}^- \\ T_{-+}^- \\ 0 \\ 0 \end{pmatrix} \frac{e^{\sqrt{s/\kappa_-}y}}{2\sqrt{s\kappa_-}}$$

- Two diagrams for $y < x_-$: II. (Laplace Wave Train Picture)



$$\begin{array}{c}
 \begin{array}{c}
 U_-^{n+1} = e^{\frac{\sqrt{s}}{\kappa_-} \Delta x} T_{-+}^- U_+^n \\
 \leftarrow U_-^n
 \end{array}
 \quad
 \begin{array}{c}
 U_+^{n+1} = e^{\frac{\sqrt{s}}{\kappa_+} \Delta x} R_{-+}^+ U_-^n \\
 S_-^{n+1} = e^{\frac{\sqrt{s}}{\kappa_-} \Delta x} R_{++}^- U_+^n \\
 \leftarrow U_+^n
 \end{array}
 \quad
 \begin{array}{c}
 S_+^{n+1} = e^{\frac{\sqrt{s}}{\kappa_+} \Delta x} T_{+-}^+ U_-^n \\
 \leftarrow S_+^n
 \end{array}
 \end{array}$$

$\Delta x \equiv x_+ - x_-$

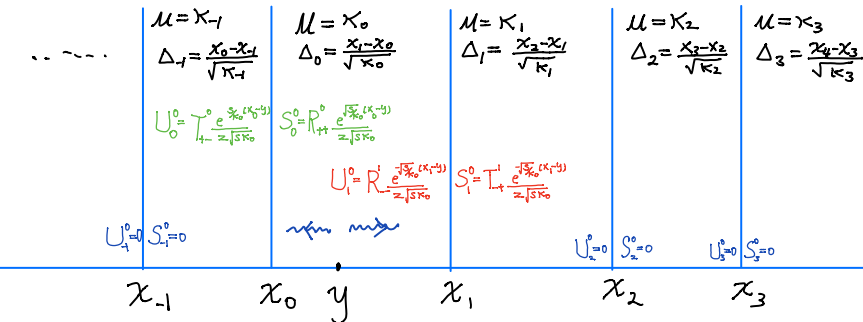
$x_- \qquad \qquad \qquad x_+$

$$\left\{ \begin{array}{l}
 U_- = \sum_{n=0}^{\infty} U_-^n = \sum_{n=0}^{\infty} (T_{-+}^- R_{-+}^+ T_{+-}^-)^n e^{\frac{\sqrt{s}}{2\sqrt{s\kappa_-}} \left(\frac{y}{\sqrt{\kappa_-}} - 2n \frac{\Delta x}{\sqrt{\kappa_0}} \right)}, \\
 \mathbb{L}[u](x, s) = \frac{e^{-\sqrt{s/\kappa_-}|x-y|}}{2\sqrt{s\kappa_-}} \\
 + \sum_{n=0}^{\infty} (T_{-+}^- R_{-+}^+ T_{+-}^-)^n e^{\frac{\sqrt{s}}{2\sqrt{s\kappa_-}} \left(\frac{y}{\sqrt{\kappa_-}} - 2n \frac{\Delta x}{\sqrt{\kappa_0}} + \frac{(|x-x_-|)}{\sqrt{\kappa_+}} \right)} \text{ for } x, y < x_-
 \end{array} \right.$$

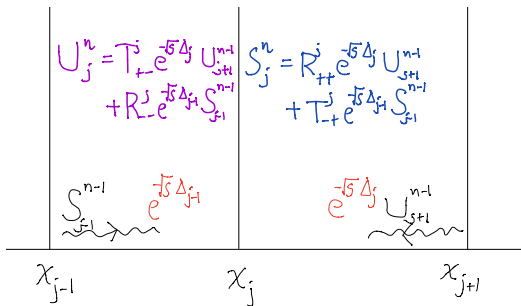
$$u(x, t) = \frac{e^{-\frac{(x-y)^2}{4\kappa_- t}}}{\sqrt{4\pi\kappa_-}} + \sum_{n=0}^{\infty} (T_{-+}^- R_{-+}^+ T_{+-}^-)^n e^{-\frac{\left(\frac{|x-x_-| + |y-x_-|}{\sqrt{\kappa_-}} + 2n \frac{\Delta x}{\sqrt{\kappa_0}} \right)^2}{4t}}}{\sqrt{4\pi\kappa_-}} \text{ for } x, y < x_-$$

Composite wave train III

$$\begin{cases} \partial_t u = \partial_x \mu(x) \partial_x u, & \mu(x) = \kappa_j \text{ for } x \in (x_j, x_{j+1}), \\ u(x, 0) = \delta(x - y), \end{cases}$$

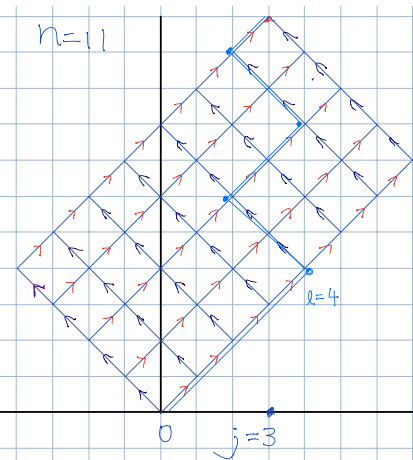


The diagram and the difference scheme



$$\begin{pmatrix} U_j^n \\ S_j^n \end{pmatrix} = \underbrace{\begin{pmatrix} T_{+-}^j & 0 \\ R_{++}^j & 0 \end{pmatrix} e^{-\sqrt{s}\Delta_j}}_{\mathcal{L}_{j+1}: \text{move to left}} \begin{pmatrix} U_{j+1}^{n-1} \\ S_{j+1}^{n-1} \end{pmatrix} + \underbrace{\begin{pmatrix} 0 & R_{--}^j \\ 0 & T_{-+}^j \end{pmatrix} e^{-\sqrt{s}\Delta_{j-1}}}_{\mathcal{R}_{j-1}: \text{move to right}} \begin{pmatrix} U_{j-1}^{n-1} \\ S_{j-1}^{n-1} \end{pmatrix}$$

Sample paths, composite Laplace wave train III, and path integral

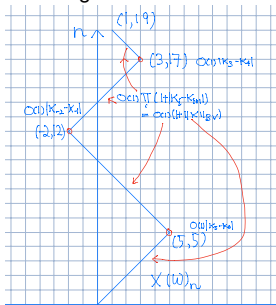


$$\left\{ \begin{array}{l} \Omega^n \equiv \{-1, 1\}^n \equiv \{\{a_i\}_{i=1}^n \mid a_i \in \{-1, 1\}\} \\ \omega \in \Omega^n: X(\omega)_k \equiv \sum_{i=1}^k \omega_i: \text{A path from 0} \\ \Omega_j^n \equiv \{\omega \in \Omega^n \mid \sum_{i=1}^n \omega_i = j\}, \\ \Omega_{j,l}^n \equiv \{\omega \in \Omega_j^n \mid \text{The path } X(\omega) \text{ changes direction } l \text{ times}\}, \\ \Psi(\omega) \equiv - \sum_{i=0}^{n-1} \frac{|x_{X(\omega)_{i+1}} - x_{X(\omega)_i}|}{\sqrt{\kappa} X(\omega)_i}: \text{phase length} \\ \Psi_+(\omega) \equiv - \sum_{i=0}^{n-1} \frac{|x_{X(\omega)_{i+1}+1} - x_{X(\omega)_i+1}|}{\sqrt{\kappa} X(\omega)_{i+1}}, \\ \mathcal{D}(\omega) = D_1 D_2 \cdots D_n, \mathcal{D}^+(\omega) = D_1^+ D_2^+ \cdots D_n^+, \\ D_i = \mathcal{L}_{X(\omega)_i} \text{ if } \omega(i) = -1, \text{ otherwise } D_i = \mathcal{R}_{X(\omega)_i}, \\ D_i^+ = \mathcal{L}_{X(\omega)_{i+1}} \text{ if } \omega(i) = -1, \text{ otherwise } D_i^+ = \mathcal{R}_{X(\omega)_{i+1}}. \end{array} \right.$$

The scattering data

$$\begin{aligned}
 & \sum_{n=0}^{\infty} \begin{pmatrix} U_j^n \\ S_j^n \end{pmatrix} \\
 &= \sum_{n=0}^{\infty} \sum_{\omega \in \Omega_j^n} \frac{\mathcal{D}(\omega) e^{\frac{\sqrt{s}(\psi(\omega) - \frac{y-x_0}{\sqrt{\kappa_0}})}}{2\sqrt{s\kappa_0}} + \sum_{n=0}^{\infty} \sum_{\omega \in \Omega_{j-1}^n} \frac{\mathcal{D}^+(\omega) e^{\frac{\sqrt{s}\psi_+(\omega) + \frac{(y-x_1)}{\sqrt{\kappa_0}}}}{2\sqrt{s\kappa_0}} \\
 &= \sum_{l=0}^{\infty} \left(\sum_{n=0}^{\infty} \sum_{\omega \in \Omega_{j,l}^n} \frac{\mathcal{D}(\omega) e^{\frac{\sqrt{s}(\psi(\omega) - \frac{y-x_0}{\sqrt{\kappa_0}})}}{2\sqrt{s\kappa_0}} + \sum_{n=0}^{\infty} \sum_{\omega \in \Omega_{j-1,l}^n} \frac{\mathcal{D}^+(\omega) e^{\frac{\sqrt{s}(\psi_+(\omega) + \frac{(y-x_1)}{\sqrt{\kappa_0}})}}{2\sqrt{s\kappa_0}} \right).
 \end{aligned}$$

Strength Estimates



$$|\mathcal{D}| \leq O(1)(1 + \|\mu\|_{BV})^{l+2} \prod_{m=1}^l |\kappa_{q_m} - \kappa_{q_{m+1}}|$$

q_m is the turning index.

$$\begin{aligned}
 & \sum_{n=0}^{\infty} \sum_{\omega \in \Omega_{j,l}^n} |\mathcal{D}| \\
 & \leq O(1)(1 + \|\mu\|_{BV})^{l+2} \sum_{q_m \in \mathbb{Z}} \prod_{m=1}^l |\kappa_{q_m} - \kappa_{q_{m+1}}| \\
 & \leq O(1)(1 + \|\mu\|_{BV})^{l+2} \sum_{q_m \in \mathbb{Z}} |\kappa_{q_m} - \kappa_{q_{m+1}}| \\
 & \leq O(1)(1 + \|\mu\|_{BV})^{l+2} \|\mu\|_{BV}^l
 \end{aligned}$$

Phase length estimate and space-time structure

- The phase length $\Psi(\omega) < -\frac{1}{2} \int_{x_1}^{x_j} \frac{dx}{\sqrt{\mu(x)}}$.
- The fundamental solution solution. $x \in (x_{j-1}, x_j)$

$$\mathbb{L}[u](x, s) = \sum_{n=0}^{\infty} \left(S_{j-1}^n e^{\sqrt{\frac{s}{\kappa_{j-1}}}(x_{j-1}-x)} + U_j^n e^{\sqrt{\frac{s}{\kappa_{j-1}}}(x-x_j)} \right)$$

$$= \sum_{\text{all possible finite paths } X(\omega) \text{ from } y \text{ to } x} C(\omega) \frac{e^{\sqrt{s}\mathcal{J}(\omega)}}{\sqrt{s}}$$

$$\left\{ \begin{array}{l} \sum_{\text{all possible finite paths } X(\omega) \text{ from } y \text{ to } x} |C(\omega)| < O(1 + \|\mu\|_{BV}), \\ \mathbb{L}^{-1}\left[\frac{e^{\sqrt{s}\mathcal{J}(\omega)}}{2\sqrt{s}}\right] = \frac{e^{-\frac{|\mathcal{J}(\omega)|^2}{4t}}}{\sqrt{4\pi t}} \leq O(1) \frac{e^{-\frac{\left(\int_x^y \frac{d\tau}{\sqrt{\mu(\tau)}}\right)^2}{8t}}}{\sqrt{4\pi t}} \end{array} \right. \Rightarrow$$

A CLASSICAL DESCRIPTION OF WEAK SOLUTION

$$|\partial_x u| \leq O(1) \frac{e^{-\frac{\left(\int_x^y \frac{d\tau}{\sqrt{\mu(\tau)}}\right)^2}{8t}}}{t}, \quad |\partial_{xy} u| \leq O(1) \frac{e^{-\frac{\left(\int_x^y \frac{d\tau}{\sqrt{\mu(\tau)}}\right)^2}{8t}}}{t^{3/2}}, \dots$$

Weak Limits

$$\begin{cases} H(x, t; y; \mu^k) \equiv u(x, t), \\ \partial_t u - \partial_x \mu^k(x) \partial_x u = 0, \\ u(x, 0) = \delta(x - y), \\ \mu^k(x): \text{A step functions.} \end{cases}$$

$$\lim_{k \rightarrow \infty} \|\mu^k - \mu\|_{\infty} = 0 \implies$$

$\lim_{k \rightarrow \infty} H(x, t; y, \mu^k) : \text{a weak solution of } (\partial_t - \partial_x \mu \partial_x) u = 0$

$$|\partial_x H(x, t; y, \mu)|, |\partial_y H(x, t; y, \mu)| \leq C_* \frac{e^{-\frac{(x-y)^2}{C_* t}}}{t},$$

$$|\partial_x \partial_y H(x, t; y, \mu)| \leq C_* \frac{e^{-\frac{(x-y)^2}{C_* t}}}{t^{3/2}}, \quad C_* > 4 \inf \mu.$$

- The Green's function $H(x, t; y, \mu)$ can be improved to $H(x, t; y, \tau; \mu)$ with $\mu = \mu(x, t)$ so that the standard iteration works.

$H(x, t; y, \tau; \mu)$: The Green's function for space-time dependent problem with a BV-function μ .

The local existence of weak solution

- The standard iteration

$$\begin{cases} \partial_t v^n - \partial_x u^n = 0, \\ \partial_t u^n + \partial_x p(v^{n-1}) - \partial_x \frac{\kappa}{v^{n-1}} \partial_x u^n = 0, \\ u^n(x, 0) = u_0(x), \quad v^n(x, 0) = 1 + v_0(x). \end{cases} \quad (u_0 \text{ and } v_0 \text{ are BV step functions}).$$

$$\mathfrak{D} \equiv \{x \in \mathbb{R} \mid (v_0, u_0)|_{x-}^{x+} \neq (0, 0)\}.$$

- Four identities.

$$\begin{aligned} (U^{n+1} - U^n)_x(x, t) &= \int_{\mathbb{R}} \left(H_x(x, t; y, 0; \kappa/V^n) - H_x(x, t; y, 0; \kappa/V^{n-1}) \right) u_0(y) dy \\ &\quad - \int_{\mathbb{R}} \int_0^t \left((H_x(x, t; y, \tau; \kappa/V^n) - H_x(x, t; y, \tau; \kappa/V^{n-1})) p(V^n)_y(y, t) \right. \\ &\quad \left. - H_x(x, t; y, \tau; \kappa/V^{n-1}) (p(V^n) - p(V^{n-1}))_y(y, t) \right) dy d\tau \\ &\quad - \sum_{z \in \mathfrak{D}} \int_0^t \left(H_x(x, t; z, \tau; \kappa/V^n) - H_x(x, t; z, \tau; \kappa/V^{n-1}) \right) p(V^n)(\cdot, t) \Big|_{z-}^{z+} d\tau \\ &\quad - \sum_{z \in \mathfrak{D}} \int_0^t H_x(x, t; z, \tau; \kappa/V^{n-1}) (p(V^n) - p(V^{n-1}))(\cdot, t) \Big|_{z-}^{z+} d\tau \\ &\quad - \int_0^t \int_{\mathbb{R} \setminus \mathfrak{D}} \int_{\tau}^t \left(H_{xy}(x, t; y, \tau; \kappa/V^n) - H_{xy}(x, t; y, \tau; \kappa/V^{n-1}) \right) (p'(V^n) U_y^n)(y, \sigma) d\sigma dy d\tau \\ &\quad - \int_0^t \int_{\mathbb{R} \setminus \mathfrak{D}} \int_{\tau}^t H_{xy}(x, t; y, \tau; \kappa/V^{n-1}) (p'(V^n) U_y^n - p'(V^{n-1}) U_y^{n-1})(y, \sigma) d\sigma dy d\tau. \end{aligned}$$

$$\begin{aligned} \|(V^{n+1} - V^n)(\cdot, t)\|_{L^1} &= \int_{\mathbb{R}} \left| \int_0^t (U^{n+1} - U^n)_x(x, \tau_1) d\tau_1 \right| dx \\ &= \kappa \int_{\mathbb{R}} \left| \int_0^t \int_{\mathbb{R} \setminus \mathcal{D}} \int_{\tau}^t H_{xy}(x, \tau_1; y, \tau; \kappa/V^n) \left(p(V^{n-1}) - p(V^n) + \frac{(V^n - V^{n-1})U_y^n}{(1+V^n)(1+V^{n-1})} \right) d\tau_1 dy d\tau \right| dx, \end{aligned}$$

$$\begin{aligned} \int_{\mathbb{R} \setminus \mathcal{D}} |(V_x^{n+1} - V_x^n)(\cdot, t)| dx &= \\ \kappa \int_{\mathbb{R} \setminus \mathcal{D}} \left| \int_0^t \int_{\mathbb{R} \setminus \mathcal{D}} \int_{\tau}^t H_{xy}(x, \tau_1; y, \tau; \kappa/V^n) \left(p(V^n) - p(V^{n-1}) + \frac{(V^n - V^{n-1})U_y^n}{(1+V^n)(1+V^{n-1})} \right) d\tau_1 dy d\tau \right| dx. \end{aligned}$$

$$\begin{aligned} \partial_t(V^{n+1} - V^n)(\cdot, t) \Big|_{x=z-}^{x=z+} &= \frac{(V^n - V^{n-1})}{\kappa}(\cdot, t) \Big|_{x=z-}^{x=z+} \left(\kappa \frac{U_x^{n+1}}{V^n} - p(V^n) \right) \\ &+ \frac{V^{n-1}}{\kappa}(\cdot, t) \Big|_{x=z-}^{x=z+} \left(\kappa \frac{U_x^{n+1}}{V^n} - \kappa \frac{U_x^n}{V^{n-1}} - p(V^n) + p(V^{n-1}) \right) + \frac{1}{\kappa} \left(V^n p(V^n) - V^{n-1} p(V^{n-1}) \right)(\cdot, t) \Big|_{x=z-}^{x=z+}. \end{aligned}$$

The metric space

$$\begin{cases} \|f\|_\infty \equiv \sup_{\sigma \in (0, t_\sharp)} \|f(\cdot, \sigma)\|_\infty, \\ \|f\|_{BV} \equiv \sup_{\sigma \in (0, t_\sharp)} \|f(\cdot, \sigma)\|_{BV}, \\ \|f\|_{L^1} \equiv \sup_{\sigma \in (0, t_\sharp)} \|f(\cdot, \sigma)\|_{L^1}, \end{cases} \quad \|f\|_{\mathfrak{D}} \equiv \sup_{\sigma \in (0, t_\sharp)} \sup_{z \in \mathfrak{D}} \left| \frac{f(\cdot, \sigma)|_{z_-}^{z_+}}{v_0(\cdot)|_{z_-}^{z_+}} \right|.$$

- The estimates

$$|||\sqrt{t}U_x^{n+1}|||_\infty \leq O(1)\delta,$$

$$\begin{aligned} ||| \frac{\sqrt{t}}{|\log t|} (U^{n+1} - U^n)_x |||_\infty &\leq O(1) \delta \left(||| V^n - V^{n-1} |||_\infty + \frac{||| V^n - V^{n-1} |||_{BV}}{|\log t_\sharp|} \right. \\ &\quad \left. + \sqrt{t_\sharp} ||| V^n - V^{n-1} |||_{L^1} + t_\sharp ||| \frac{\sqrt{t}}{|\log t|} (U^n - U^{n-1})_x |||_\infty \right), \end{aligned}$$

$$\|V^{n+1} - V^n\|_{l^1} \leq O(1)\sqrt{t_\#}(1 + O(1)\delta)\|V^n - V^{n-1}\|_{l^1},$$

$$\int_{\mathbb{R} \setminus \mathfrak{D}} |(V^{n+1} - V^n)_x(x, t)| dx \leq O(1) \|V^n - V^{n-1}\|_{L^1} (\sqrt{t} + \|\sqrt{t} U_x^n\|_\infty),$$

$$\begin{aligned} |||V^{n+1} - V^n|||_{\mathfrak{D}} &\leq O(1)\delta\sqrt{t_{\sharp}}|||V^n - V^{n-1}|||_{\mathfrak{D}} + O(1)|\log t_{\sharp}|\sqrt{t_{\sharp}}|||\frac{\sqrt{t}}{|\log t|}(U^n - U^{n-1})_x|||_{\infty} \\ &\quad + O(1)\sqrt{t_{\sharp}}|||\sqrt{t}U_x^{n-1}|||_{\infty}|||V^n - V^{n-1}|||_{\infty}. \end{aligned}$$

Standard Iteration scheme converges for $t \leq t_{\#} \ll 1$.

Conclusion I. Local existence

$(\mathbf{B}_\delta, d)$: Metric space for initial data.

$$\begin{cases} \mathbf{B}_\delta \equiv \{(v_0, u_0) : \|v_0 - 1\|_{L^1} + \|v_0\|_{BV} + \|u_0\|_{L^1} + \|u_0\|_{BV} \leq \delta, \\ d((v_0^a, u_0^a), (v_0^b, u_0^b)) \equiv \|v_0^a - v_0^b\|_{L^1} + \|u_0^a - u_0^b\|_{BV}. \end{cases}$$

$\mathbf{S}^{t_\#} \equiv \{(v_0, u_0) \in \mathbf{B}_\delta : \text{The initial data admits solution for } t \in (0, t_\#)\}$

Theorem. There exist δ and $t_\# > 0$ such that $\mathbf{S}^{t_\#}$ is dense in $(\mathbf{B}_\delta, d)$.

Hadamard's well-posedness problem

- One needs to require $0 < \gamma < e$ for $p(v) = v^{-\gamma}$.

- Green's function $\mathbb{G}(x, t; \alpha, \beta)$, $\alpha, \beta > 0$, of a linear system:

$$\begin{pmatrix} \partial_t & -\partial_x \\ -\beta^2 \partial_x & \partial_t - \alpha \partial_x^2 \end{pmatrix} \mathbb{G}(x, t; \alpha, \beta) = \vec{0}, \quad \mathbb{G}(x, 0; \alpha, \beta) = \delta(x) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

- Singular-Regular decomposition of $\mathbb{G}(x, t; \alpha, \beta)$:

$$\begin{cases} \mathbb{G} \equiv \mathbb{G}^* + \mathbb{G}^\sharp, \\ \mathbb{G}^* = \mathbb{G}^{*,1} + \mathbb{G}^{*,2}, \end{cases}$$

$$\begin{cases} \mathbb{G}^* = \mathbb{G}^{*,1} + \mathbb{G}^{*,2}, \\ \left| \partial_t^i \left(\left(\mathbb{G}^{*,1} - \delta(x) e^{-\frac{\beta^2}{\alpha} t} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \right)_x + \frac{e^{-\frac{\beta^2}{\alpha} t}}{\alpha} \delta(x) \begin{pmatrix} 0 & 1 \\ \beta^2 & 0 \end{pmatrix} \right) \right| \leq O(1) e^{-\sigma_* |x|}, \\ \mathbb{G}^{*,2}(x, t; \alpha, \beta) = O(1) e^{-\sigma^* t - \sigma_0 |x|} + \begin{cases} \frac{1}{\sqrt{4\pi\alpha t}} e^{-\frac{x^2}{4\alpha t}} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, & 0 < t < 1; \\ 0, & t \geq 1, \end{cases} \\ |\partial_x^k \mathbb{G}^\sharp| \leq O(1) \frac{e^{-\frac{x^2}{C_*(t+1)}}}{\sqrt{(t+1)^{k+1}}} \text{ for } k = 0, 1, 2. \end{cases}$$

An integral representation of the weak solution

Fixed (x, t) and $\bar{v} \equiv v(x, t)$. Take $\bar{\alpha} = \kappa/\bar{v}$ and $\bar{\beta} = \sqrt{-p'(\bar{v})}$.

Consider $\int_0^t \int_{\mathbb{R}} \mathbb{G}(x-y, t-\tau; \bar{\alpha}, \bar{\beta}) \left(\frac{\partial_\tau v - \partial_y u}{\partial_\tau u + \partial_y p(v) - \kappa \partial_y \frac{\partial_y u}{v}} \right) dy d\tau = \vec{0}$.

$$\begin{aligned} \bar{v} = & \int_{\mathbb{R}} \mathbb{G}_{11}(x-y, \bar{\beta}, \bar{\alpha}) v_0(y) + \mathbb{G}_{12}(x-y, \bar{\beta}, \bar{\alpha}) u_0(y) dy \\ & + \int_0^t \int_{\mathbb{R}} \partial_x \mathbb{G}_{12}(x-y, t-\tau; \bar{\alpha}, \bar{\beta}) \left(-(p(v) - p(\bar{v}) - p'(\bar{v})(v - \bar{v})) \right) (y, \tau) dy d\tau \\ & + \int_0^t \int_{\mathbb{R}} \partial_x \mathbb{G}_{12}(x-y, t-\tau; \bar{\alpha}, \bar{\beta}) \left(\log v - \frac{v}{\bar{v}} \right)_\tau dy d\tau. \end{aligned}$$

$$\begin{aligned}
\bar{v} = & \left(e^{-\frac{\bar{\beta}^2}{\bar{\alpha}} t} \left(\bar{v} \log \frac{v_0(x)}{\bar{v}} + \bar{v} - v_0(x) \right) + \int_{\mathbb{R}} \mathbb{G}_{11}(x-y, t; \bar{\alpha}, \bar{\beta}) v_0(y) dy \right. \\
& \left. + \int_{\mathbb{R}} \mathbb{G}_{12}(x-y, t; \bar{\alpha}, \bar{\beta}) u_0(y) dy \right. \\
& + \kappa \int_{\mathbb{R}} \left(D_{12}(x-y, 0; \bar{\alpha}, \bar{\beta}) \left(\log \left(\frac{v(y, t)}{\bar{v}} \right) - \frac{v(y, t) - \bar{v}}{\bar{v}} \right) - D_{12}(x-y, t; \bar{\alpha}, \bar{\beta}) \left(\log \left(\frac{v_0(y)}{\bar{v}} \right) - \frac{v_0(y) - \bar{v}}{\bar{v}} \right) \right) dy \\
& + \int_0^t \frac{1}{\bar{\alpha}} e^{-\frac{\bar{\beta}^2}{\bar{\alpha}}(t-\tau)} (p(v(x, \tau)) - p(\bar{v}) - p'(\bar{v})(v(x, \tau) - \bar{v})) d\tau \\
& + \int_0^t \int_{\mathbb{R}} \partial_y (\mathbb{G}_{12} - \mathbb{G}_{12}^{*,1}) (x-y, t-\tau; \bar{\alpha}, \bar{\beta}) (p(v(y, \tau)) - p(\bar{v}) - p'(\bar{v})(v(y, \tau) - \bar{v})) dy d\tau \\
& - \int_0^t \int_{\mathbb{R}} \partial_y (\mathbb{G}_{12} - \mathbb{G}_{12}^{*,1}) (x-y, t-\tau; \bar{\alpha}, \bar{\beta}) \kappa \left(\frac{u_y(y, \tau)}{v(y, \tau)} - \frac{u_y(y, \tau)}{\bar{v}} \right) dy d\tau \\
& + \int_0^t \frac{\bar{\beta}^2}{\bar{\alpha}} e^{-\frac{\bar{\beta}^2}{\bar{\alpha}}(t-\tau)} \left(\bar{v} \log \left(\frac{v(x, \tau)}{\bar{v}} \right) - (v(x, \tau) - \bar{v}) \right) d\tau \\
& + \int_0^t \int_{\mathbb{R}} O(1) e^{-\sigma_*(t-\tau) - \sigma_0|x-y|} \left(\log \left(\frac{v(y, \tau)}{\bar{v}} \right) - \frac{v(y, \tau) - \bar{v}}{\bar{v}} \right) dy d\tau.
\end{aligned}$$

$$\begin{aligned}
u_x = & \int_{\mathbb{R}} H_x(x, t; y, 0; \kappa/v) u_0(y) dy - \int_0^t \int_{\mathbb{R}} H_x(x, t; y, \tau; \kappa/v) p(v(y, t))_y dy d\tau \\
& - \sum_{z \in \mathfrak{D}} \int_0^t H_x(x, t; z, \tau; \kappa/v) p(v(t), d\tau|_{z^-}^{z^+}) + \int_0^t \int_{\mathbb{R} \setminus \mathfrak{D}} \int_{\tau}^t H_{xy}(x, t; y, \sigma; \kappa/v) (p'(v(y, \sigma)) u_y(y, \sigma)) d\sigma dy d\tau.
\end{aligned}$$

$$\sup_{t>0} \frac{d}{d\bar{v}} e^{-\frac{\beta^2}{\alpha} t} (\bar{v} \log \frac{v_0(x)}{\bar{v}} + \bar{v}) = \sup_{t>0} \frac{d}{d\bar{v}} e^{\frac{\rho'(\bar{v})\bar{v}}{\kappa} t} (\bar{v} \log \frac{v_0(x)}{\bar{v}} + \bar{v}) < \gamma/e < 1 \text{ if } \gamma < e.$$

$\Rightarrow$

Theorem. For any two initial data (v_0^a, u_0^a) and $(v_0^b, u_0^b) \in \mathbf{S}^{t\sharp}$, the solutions (v^a, u^a) and (v^b, u^b) satisfy

$$\|(v^a - v^b)(\cdot, t)\|_{L^1} + \|(u^a - u^b)(\cdot, t)\|_{L^1} \leq O(1) (\|v_0^a - v_0^b\|_{L^1} + \|u_0^a - u_0^b\|_{L^1}).$$

Hadamard's Well-Posedness problem for NS in $\|\cdot\|_{BV} + \|\cdot\|_{L^1}$ is **SOLVED**.